RAJALAKSHMI ENGINEERING COLLEGE

Dharanipriyan S 251801056-2025-AIDS

D2

Started on	Thursday, 24 September 2026, 8:18 AM
State	Finished
Completed on	Thursday, 24 September 2026, 8:48 AM
Time taken	30 mins 9 secs
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 10.00 out of 10.00

Playing with Numbers:

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3.Write any efficient algorithm to find the possible ways.

Example 1:

Input: *6*

Output:*6*

Explanation: *There are 6 ways to 6 represent number with 1 and 3*

1+1+1+1+1+1

3+3

1+1+1+3

1+1+3+1

1+3+1+1

3+1+1+1

Input Format

First Line contains the number n

Output Format

Print: The number of possible ways 'n' can be represented using 1 and 3

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int main() {
    int n;
    scanf("%d", &n);

    if (n < 0) {
        printf("0\n");
        return 0;
    }
    if (n == 0 || n == 1 || n == 2) {
        printf("1\n");
        return 0;
    }

    unsigned long long ways[n + 1];

    ways[0] = 1;
    ways[1] = 1;
    ways[2] = 1;

    for (int i = 3; i <= n; i++) {
        ways[i] = ways[i - 1] + ways[i - 3];
    }

    printf("%llu\n", ways[n]);
```

```
26      printf("%lld\n", ways[n]);
27
28      return 0;
29  }
30
```

	Input	Expected	Got	
✔	6	6	6	✔
✔	25	8641	8641	✔
✔	100	24382819596721629	24382819596721629	✔

Passed all tests! ✔

Correct

Marks for this submission: 10.00/10.00.

Back to Course